

Solutions to Problem Set 3

1. A speech signal is sampled at a rate of 8 kHz, and then encoded using 8 bits per sample. The binary stream is then transmitted through an AWGN baseband channel via M-level PAM (Pulse Amplitude Modulation). The ISI (Inter-Symbol Interference) should be eliminated.

- a) Determine the minimum bandwidth required for transmission when $M = 4$.
- b) Determine the minimum bandwidth required for transmission when $M = 16$. What is the price to pay for saving the bandwidth?

Solution. a) The data rate of the binary stream is $R_b = 8\text{kHz} \times 8\text{bits} = 64\text{kbps}$.

For M-level MPAM, each symbol can carry $\log_2 M$ bits. When $M = 4$, the symbol rate is $R_s = R_b / \log_2 M = 32\text{kHz}$. According to the Nyquist criterion, the minimum bandwidth satisfies $2W \leq R_s$, with equality if and only if the spectrum of the impulse waveform is an ideal low-pass filter. Therefore,

$$W \leq R_s/2 = 16\text{kHz}$$

- b) When $M = 16$, the symbol rate is $R_s = R_b / \log_2 M = 16\text{kHz}$. The minimum bandwidth is $R_s/2 = 8\text{kHz}$. When we encode more bits into one symbol to save bandwidth, if the symbol power is fixed, the symbol levels are closer to each other, thus more errors could happen.

2. For a 4-level PAM baseband transmission system, the transmitted signals consists of pulses having a raised cosine spectrum with a roll-off factor 0.5 and its spectrum extends to 7.5kHz.

a) What is the maximum data rate with zero-ISI requirement? What is the corresponding maximum bandwidth efficiency?

b) Will there be any ISI when the symbol period is 0.2ms? What about 0.15ms?

Explain in the perspectives of both the time domain and the frequency domain.

Solution. a) For a given raised cosine spectrum, the maximum symbol rate with zero-ISI is

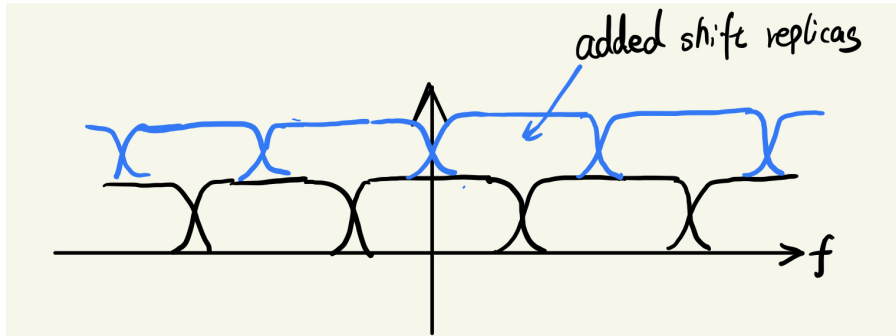
$$R_{s,\max} = \frac{2W}{\alpha + 1} = 10\text{kHz}$$

Thus the maximum data rate and the maximum bandwidth efficiency are

$$R_{b,\max} = R_{s,\max} \log_2 M = 20\text{kbps}$$

$$\eta_{\max} = R_{b,\max}/W = \frac{8}{3} = 2.67\text{bps/Hz}$$

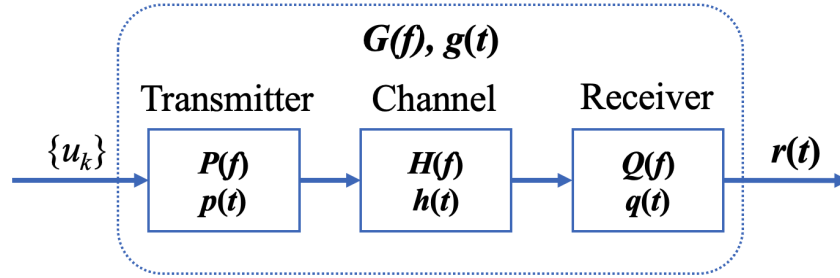
b) If the $T_s = 0.2\text{ms}$, then $R_s = 5\text{kHz} = \frac{1}{2}R_{s,\max}$. In the time domain, when the symbol period is doubled from the minimum symbol rate, the zero-ISI property would be preserved, because given $g(kT_s) = 0, k \neq 0$, we have $g(k \cdot 2T_s) = 0, k \neq 0$. In the frequency domain, when the symbol rate is halved, the shifted replicas will be denser but the sum is still a constant, as shown in the figure below. In fact, when the symbol period is a multiple of the minimum symbol period with zero ISI, then it has zero ISI as well.



However, if the $T_s = 0.15\text{ms}$, in the time domain $g(k \cdot 1.5T_s), k \neq 0$ is not guaranteed to be zero, while in the frequency domain the sum of shifted replicas is not constant for all frequencies.

To conclude, $T_s = 0.2\text{ms}$ results in no ISI but $T_s = 0.15\text{ms}$ will have ISI.

3. Consider a PAM baseband transmission system. The baseband waveform is $p(t)$ with the symbol interval T_s , the channel is characterized by a filter $h(t)$, and the filter at the receiver is $q(t)$. As a result, the received waveform is given by $r(t) = \sum_k u_k g(t - kT)$, where $g(t) = p(t) * h(t) * q(t)$ and u_k is the sequence of symbols.



- a) What property must $g(t)$ have, so that $r(kT) = u_k$ for any k ?
- b) Recall the Fourier transform of $\text{sinc}(x)$ is

$$\mathcal{F}\{\text{sinc}(x)\} = \text{rect}(f) = \begin{cases} 1, & f \in [-0.5, 0.5] \\ 0, & \text{else} \end{cases}.$$

Let the transmitter waveform be $p(t) = \text{sinc}(1000t)\text{sinc}(2000t)$. Calculate the frequency domain representation $P(f)$ of the waveform.

- c) Assume the symbol time is $T_s = 0.5\text{ms}$, and the frequency response of the channel is specified by

$$H(f) = \begin{cases} 1, & |f| \in [0, 0.75\text{kHz}] \\ 0, & |f| \in (0.75\text{kHz}, 1\text{kHz}] \\ 1, & |f| \in (1\text{kHz}, 1.25\text{kHz}] \\ 0, & |f| \in (1.25\text{kHz}, \infty] \end{cases}.$$

Can you design a receiver filter $Q(f)$ so that there is no ISI?

(Hint: The spectrum $G(f)$ of $g(t)$ should satisfy the condition that $G(f) + G(1/T_s - f)$ is a constant, according to the Nyquist criterion)

- d) If $H(f)$ is non-zero only at $[0, 0.75\text{kHz}]$, is it possible to design a no-ISI receiver and why?

Solution. a) $g(t)$ should have no ISI, specifically,

$$g(kT) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

b) Apply Fourier transform directly on $p(t)$,

$$\begin{aligned} P(f) &= \mathcal{F}\{p(t)\} \\ &= \mathcal{F}\{\text{sinc}(1000t)\} * \mathcal{F}\{\text{sinc}(2000t)\} \\ &= 2 \times 10^{-6} \times \text{rect}(f/1000) * \text{rect}(f/2000) \\ &= 2 \times 10^{-6} \cdot \begin{cases} 1, & |f| \in [0, 0.5\text{kHz}] \\ \frac{1500-f}{1000}, & |f| \in [0.5\text{kHz}, 1.5\text{kHz}] \\ 0, & |f| \in [1.5\text{kHz}, \infty] \end{cases} \end{aligned}$$

We omit the constant term below for simplicity.

c) In order that $g(t)$ has no ISI, Nyquist criterion requires $G(f) = P(f)H(f)Q(f)$ to satisfy that $G(f) + G(2000 - f)$ is a constant. Now

$$P(f)H(f) = \begin{cases} 1, & |f| \in [0, 0.5\text{kHz}] \\ \frac{1500-f}{1000}, & |f| \in [0.5, 0.75\text{kHz}] \\ 0, & |f| \in [0.75, 1\text{kHz}] \\ \frac{1500-f}{1000}, & |f| \in [1, 1.25\text{kHz}] \\ 0, & |f| \in [1.25\text{kHz}, \infty] \end{cases}$$

Let $G(f) + G(2000 - f) = 1$. In the region $|f| \in [0, 0.75\text{kHz}]$ and $|f| \in [1\text{kHz}, 1.25\text{kHz}]$,

$$G(f) + G(2000 - f) = G(f) + 0 = P(f)H(f)Q(f) = 1$$

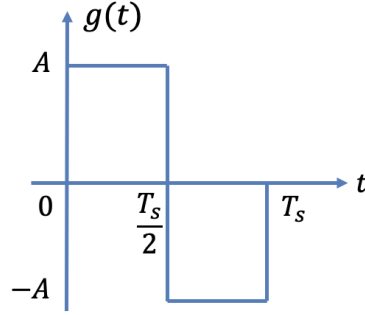
thus

$$Q(f) = \begin{cases} 1, & |f| \in [0, 0.5\text{kHz}] \\ \frac{1000}{1500-f}, & |f| \in [0.5\text{kHz}, 0.75\text{kHz}] \\ \frac{1000}{1500-f}, & |f| \in [1\text{kHz}, 1.25\text{kHz}] \end{cases}$$

Since $P(f)H(f) = 0$ in other frequencies, the value of $Q(f)$ is not important.

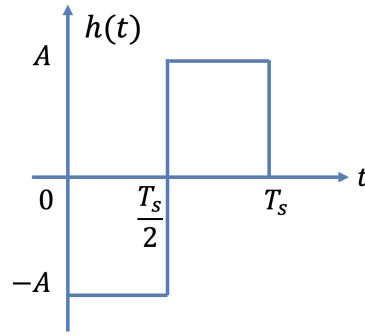
d) For $H(f)$ constrained to $[0, 0.75\text{kHz}]$, $G(f) + G(2000 - f) = 0$, $f \in [0.75\text{kHz}, 1.25\text{kHz}]$, which violates the Nyquist criterion. Therefore it is impossible to design a no-ISI receiver.

4. In a baseband transmission system, the output waveform of the transmitter is given below. The channel is additive white Gaussian noise (AWGN) channel with noise PSD $n_0/2$.



- Design a matched filter for the optimal receiver and draw the impulse response of the filter.
- Draw the output of the matched filter.
- Compute the optimal SNR at the sampled output after the matched filter.

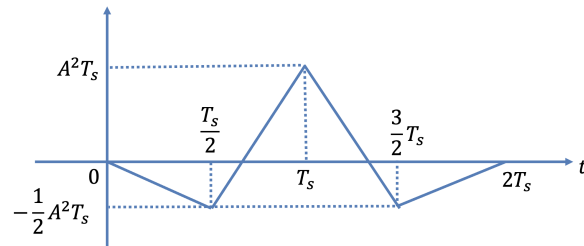
Solution. a) The matched filter is $h(t) = g(T_s - t)$, as shown below.



- b) The output of the matched filter is

$$\begin{aligned} g(t) * h(t) &= \int g(\tau)h(t - \tau)d\tau \\ &= \int g(\tau)g(\tau + T_s - t)d\tau \end{aligned}$$

which can be calculated by sweeping the value of t . The result is shown as the figure below.



c) The output signal is sampled at time T_s for optimal SNR. The energy of the sampled signal output is $g_o^2(T_s) = (A^2 T_s)^2$. On the other hand, the white Gaussian noise goes through the matched filter, thus $N_o(f) = \frac{n_0}{2} |G(f)|^2$. The variance of the noise is

$$\begin{aligned}
\mathbb{E}[n_o(t)^2] &= \mathcal{F}^{-1}[N_o(f)](0) \\
&= \int_{-\infty}^{\infty} \frac{n_0}{2} |G(f)|^2 e^{j2\pi 0 f} df \\
&= \frac{n_0}{2} \int_{-\infty}^{\infty} |g(t)|^2 dt \\
&= \frac{n_0}{2} A^2 T_s
\end{aligned}$$

Therefore, the SNR is $g_o^2(T_s)/\mathbb{E}[n_o(t)^2] = 2A^2 T_s/n_0$.

5. In this problem, we derive the matched filter in the time domain.

Suppose a symbol a_i is modulated on a baseband pulse waveform $g(t)$. The receiver filter is $h(t)$ and is sampled at $t = T_s$. Consider an additive white Gaussian noise $n(t)$ with PSD (Power Spectral Density) $n_0/2$. The sampled signal is:

$$Y = S + N,$$

where $S = a_i \int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau$, $N = \int_{-\infty}^{\infty} n(\tau)h(T_s - \tau) d\tau$.

a) Prove the expression of the Signal-to-Noise Ratio (SNR) at the output sample point is

$$\text{SNR} = \frac{S^2}{\mathbb{E}[N^2]} = \frac{2a_i^2}{n_0} \frac{\left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\int_{-\infty}^{\infty} h^2(\tau) d\tau}$$

b) Find the maximum SNR and the corresponding $h(t)$.

(Hint: use the Cauchy-Schwarz inequality: $(\int f(x)g(x) dx)^2 \leq \int f^2(x) dx \int g^2(x) dx$, with equality if and only if $g(x) = kf(x)$)

Solution. a) The SNR at the output sample point is

$$\begin{aligned} \text{SNR} &= \frac{S^2}{\mathbb{E}[N^2]} = \frac{a_i^2 \left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\mathbb{E} \left\{ \left[\int_{-\infty}^{\infty} n(\tau)h(T_s - \tau) d\tau \right]^2 \right\}} \\ &= \frac{a_i^2 \left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\mathbb{E} \left\{ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} n(\tau)h(T_s - \tau)n(\tau')h(T_s - \tau') d\tau d\tau' \right\}} \end{aligned}$$

Since the noise is white, i.e. $\mathbb{E}[n(t)n(s)] = \frac{n_0}{2}\delta(t - s)$,

$$\begin{aligned} \text{SNR} &= \frac{a_i^2 \left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\tau - \tau')h(T_s - \tau)h(T_s - \tau') d\tau d\tau'} \\ &= \frac{a_i^2 \left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{\frac{n_0}{2} \int_{-\infty}^{\infty} h(T_s - \tau)h(T_s - \tau) d\tau} \\ &= \frac{2a_i^2 \left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau) d\tau \right]^2}{n_0 \int_{-\infty}^{\infty} h^2(\tau) d\tau} \end{aligned}$$

b) By Cauchy-Schwarz inequality,

$$\begin{aligned} \text{SNR} &\leq \frac{2a_i^2}{n_0} \frac{\int_{-\infty}^{\infty} g^2(\tau) d\tau \int_{-\infty}^{\infty} h^2(T_s - \tau) d\tau}{\int_{-\infty}^{\infty} h^2(\tau) d\tau} \\ &= \frac{2a_i^2}{n_0} \int_{-\infty}^{\infty} g^2(\tau) d\tau \end{aligned}$$

with equality if and only if $h(t) = kg(T_s - t)$.